



Final report / self-assessment report

1. Briefly document each step of the project here, including any issues encountered, solutions implemented, and insights gained from the data. This section can be collaboratively written and shared by all team members.

- Initially we created a communication method, Teams for official documentation and WhatsApp for quick and direct communication.
- Our team had an initial meeting to divide workload around the 7th of November.
- The tasks were split between team members based on the points assigned to the tasks, so that each of the team members has an approximately equal number of points under their responsibility.
- Soon after we created a GitHub repo and started working on our own branches.
- Eduard had to start first with the extract function and setting up a rough draft skeleton for the project. Decided to use the Kaggle API to extract the csv instead of just adding the csv manually in a folder. Added some extra fun validations and prints for that.
- Zoi was responsible for the Transform step as well as part of the Validation.

More specifically for:

- o Converting the column "Formatted Date" in appropriate format.
 - o Implementing the cleaning part (handling missing-invalid values, filtering out pressure measurements out of reasonable limits, and removing duplicates.
 - o Completing the feature engineering part as described in the assignment:
 - o Computing daily and monthly averages, calculating monthly precipitation mode and categorizing wind strength.
 - o Saving the modified daily and monthly files and passing their paths to next task using Xcom.
 - o Working for the first validation parts, specifically checking the checks about missing values and verification that the aggregated values fall within reasonable ranges.
 - o Designing the parallel execution structure for the validation task in Airflow by creating separate validation subtasks so that they can run in parallel and the merge through the join task.
 - o Contributing to the discussion about an inconsistency between the feature engineering and load requirements, and helping the group agree on the most logically correct solution (keeping aggregated data only).
 - o Ensuring all transform functions were modular and easy for the downstream tasks.
- Iuliia was responsible for
 - o Creating tables in the database and loading data into them; for this, the PythonOperator was used when defining the task. Using SqliteOperator might be a better option for a real-life project, but here it was concluded that it is better to use PythonOperator and give it the path to the database file in order to avoid having to set



- up the database via Airflow. The data loading was split into two tasks: one for loading daily data and one for loading monthly data; thus, if one of the tasks fails, it can be guaranteed that the data is not duplicated.
- Maintaining Airflow and ensuring that the DAG runs successfully; this involved ensuring paths and imports are correct (in particular, the information was used that best practices for Airflow include putting most of the imports within functions, not in the top level).
 - During the process, we communicated well and on time.
 - Issues:
 - Small difficulties with merging branches and reading each other's code but all went smoothly.
 - There was an issue of inconsistency between "Feature engineering" and data loading tasks in the project description: feature engineering task mentions just averaging all data to produce daily values, but data loading tasks prescribes to also add the non-averaged data to the database table; thus, it was not clear how to proceed. It was not clear how to proceed, since mixing hourly data with daily data is likely an incorrect solution. It was agreed upon to only keep the daily average values.
 - From Eduard's part, small difficulties with the outlier validation, he wanted to make it a bit more than just "Temperature > 40", hence used lower and upper quartile to solve the problem.
 - Some insights we got from the data includes: ...
 - We had a last meeting on the day of submission and one before the presentation to check the pipeline and ensure it works smoothly.

2. Include a brief explanation of each group member's contributions and activities. Be sure to identify any students who had minimal or no participation throughout the project:

2.1. Final Grade: Enter the points earned here. The maximum points is 28 points.

28/28 points! + 2 points for using git.

2.2. Tasks completed by each member:

List the full names of all group members and provide a short description of the tasks each person performed. Here are some examples:

- **Eduard Rednic:** Responsible for the extract function, related tasks, github repo set up, structure of the scripts with comments and some first draft generic code, and outlier validation.
- **Zoi Theofilakou:** Handled the transform function, associated tasks and validations.
- **Iuliia Radionova:** Focused on loading, debugging the DAG execution in Airflow.

Note: Each student must individually submit this form to the "Teamwork Submission" assignment in Itslearning.